$$\Delta S_V = \frac{+35200\ J\ mol^{-1}}{383.6\ K}$$

$$\Delta S_V = +\ 91.76\ J\ K^{-1}\ mol^{-1}$$

60.

Solution :

Given :

$\Delta H = 30.56\ kJ\ mol^{-1}$

$= 30560\ J\ mol^{-1}$

$\Delta S = 6.66 \times 10^{-3}\ kJK^{-1}mol^{-1}$

T = ? at which $\Delta G = 0$

$\Delta G = \Delta H - T\Delta S$

$0 = \Delta H - T\Delta S$

$$T = \frac{\Delta H}{\Delta S}$$

$$T = \frac{30.56 kJ mol^{-1}}{6.66 \times 10^{-3} kJK^{-1} mol^{-1}}$$

T = 4589 K

(i) At 4589K ; $\Delta G = 0$ the reaction is in equilibrium.

(ii) at temperature below 4598 K , $\Delta H > T\Delta S$

$\Delta G = \Delta H - T\Delta S > 0$, the reaction in the forward direction, is non spontaneous. In other words the reaction occurs in the backward direction.

61.

Solution :

Given

$T = 400K ; \Delta H^0 = 77.2\ kJ\ mol^{-1} = 77200$

$J\ mol^{-1} ; \Delta S^0 = 122\ JK^{-1}\ mol^{-1}$

$\Delta G^0 = -2.303\ RT \log K_{eq}$

$$\log K_{eq} = \frac{-\Delta G^0}{2.303\ RT}$$

$$\log K_{eq} = -\frac{\left(\Delta H^0 - T\Delta S^0\right)}{2.303\ RT}$$

$$\log K_{eq} = -\left(\frac{77200 - 400 \times 122}{2.303 \times 8.314 \times 400}\right)$$

$$\log K_{eq} = -\left(\frac{28400}{7659}\right)$$

$\log K_{eq} = -\ 3.7080$

$K_{eq} = \text{antilog}\ (-3.7080)$

$K_{eq} = 1.95 \times 10^{-4}$

62.

Solution :

Given

$T = 298K ; \Delta U = -\ 742.4\ kJ\ mol^{-1}$

$\Delta H = ?$

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta H = \Delta U + (n_p - n_r)\ RT$

$$\Delta H = -\ 742.4 + \left(2 - \frac{3}{2}\right) \times 8.314 \times 10^{-3} \times 298$$

$= -\ 742.4 + (0.5 \times 8.314 \times 10^{-3} \times 298)$

$= -\ 742.4 + 1.24$

$= -\ 741.16\ kJ\ mol^{-1}$

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